

Design Verification Plan & Report (Virtual) - t6_r3

Doc No: VBF-T6R3-DVPR-01	Case: t6_r3	Rev: 01	Date: 2026-08-26
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All verification below is ****virtual**** (DFN simulation + calc-energy), executed through the bda harness and logged via log-evaluate against the immutable entry-0 thresholds. Physical DV testing is outside this virtual design's scope and remains open.

No	Test item	Requirement	Method (virtual)	Result	Margin	Verdict	Source
1	Volumetric energy density	>= 950 Wh/L	calc-energy on DFN 1C discharge	983.61 Wh/L	+33.61 Wh/L	PASS	eval r10 V27; energy json
2	Voltage plateau	>= 4.1 V	calc-energy midpoint_voltage_v	4.1551 V	+55.1 mV	PASS	eval r10 V27; energy json
3	4C fast charge, no Li plating	anode potential >= 0 V throughout	DFN 4C CC-CV (18 A), min anode_potential_v	+0.1050 V	+105.0 mV	PASS	eval r10 V27; 4c json
4	Max temperature, 4C charge	<= 323.15 K (50 C)	DFN 4C T_max_K	318.922 K	-4.23 K	PASS	eval r10 V27; 4c json
5	Anode SEI after 100 cycles	<= 500 nm	DFN aging, sei_thickness_nm_end	313.7 nm	-186.3 nm	PASS	eval r10 V27; aging json
6	DC resistance (supplemental)	record	calc-energy dcr_ohm	4.986 mOhm	-	RECORD	energy json
7	Cell mass (supplemental)	record	calc-energy mass_kg	47.6800 g dry	-	RECORD	energy json
8	0-100% SOC 4C charge, no plating	recommended	DFN with protocol starting from lower cut-off	NOT RUN	-	OPEN	protocol limitation (see note)
9	True DFT/MD endorsement	SKILL stage 5	run-orca / run-md	SKIPPED	-	SKIPPED honestly	real_compute=false, endorse entry
10	Physical build & DV testing	production gate	physical cell build	NOT RUN	-	OPEN	virtual design scope

Notes

- Row 3 note: the harness 4C protocol charges from the model initial state (cell at 4.1995 V, nearly full), so the CC phase lasts ~16.2 ms and the charge adds only 8.09e-05 Ah. The plating-free result is valid for the imposed protocol (anode never dips below +0.105 V) and is further protected by N/P 1.579; a deep-discharge-start 4C verification is tracked as row 8 OPEN. This limitation is disclosed, not hidden.
- Rows 1-5 were evaluated by `bda log-evaluate` round 10: candidate V27_combo verdict=pass, checked=5.
- Rounds 1-9 failures (incl. the round-9 DFN rejection of the SPMe finalist) are preserved in log.jsonl and are part of the audit trail.